

# Pushforward Measures and Change of Variables

## Pushforward

Let

$$F : X \rightarrow Y$$

be a measurable map and let  $\mu$  be a probability measure on  $X$ . The pushforward measure

$$F_{\#}\mu$$

is the probability measure on  $Y$  defined by

$$(F_{\#}\mu)(B) = \mu(F^{-1}(B))$$

for every measurable set  $B \subseteq Y$ .

## Defining Integral Property

The defining integral property of the pushforward is:

$$\int_Y \Phi(y) d(F_{\#}\mu)(y) = \int_X \Phi(F(x)) d\mu(x)$$

for every integrable function  $\Phi : Y \rightarrow \mathbb{R}$ .

## Why This Is a Change of Variables

The left-hand side integrates over the pushed-forward variable  $y$ . The right-hand side pulls the same function back to the original variable  $x$  by composing with  $F$ .

Thus the notation

$$y = F(x)$$

should be understood measure-theoretically as

$$d(F_{\#}\mu)(y) \text{ is the distribution of } F(x) \text{ when } x \sim \mu.$$

## Application to the Geodesic Proof

In the Wasserstein geodesic proof, we define

$$F_{s,t}(x) = (T_s(x), T_t(x))$$

and

$$\gamma_{s,t} = (F_{s,t})_{\#}\mu.$$

Taking

$$\Phi(a, b) = |a - b|^2$$

gives

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} |a - b|^2 d\gamma_{s,t}(a, b) = \int_{\mathbb{R}^d} |T_s(x) - T_t(x)|^2 d\mu(x).$$

The apparent double integral becomes a single integral because  $\gamma_{s,t}$  is not arbitrary: it is generated by one original particle label  $x$ .